DSAA3073 · Theories in Data Science

# Predict before the outcome is known

## Learning Sheet · Lecture 10

Expert advice, private sampling, and multiplicative weights

# Choose the comparison before the algorithm

**LESSON BACKBONE**

**How can we predict sequentially and stay close to one good expert without assuming that any expert is perfect?**

We first define the decisions, feedback and comparison. We then move from removing mistaken experts to reducing their weights. Finally, private sampling and two bounds on total weight give a guarantee on the extra loss.

Suppose several prediction rules offer advice each day. We must decide before seeing that day's outcome. Afterward, we can see which rules were correct and use this information tomorrow. A rule that has worked well so far may still be wrong today. The task is to combine the available advice without knowing future outcomes.

Call each advice rule an **expert**. An expert need not be a person, and the name does not promise that its advice is accurate. Our learner chooses a prediction using the experts' advice and the information already available. We will compare the learner with the best one of these experts on the same observed sequence.

This differs from asking how a fitted classifier will perform on new data from a population. Here, prediction and feedback occur repeatedly in time. The sequence need not be independent or drawn from one fixed distribution. We can still measure the actual losses on that sequence. The guarantee will concern a comparison between those losses, rather than a claim about population risk.

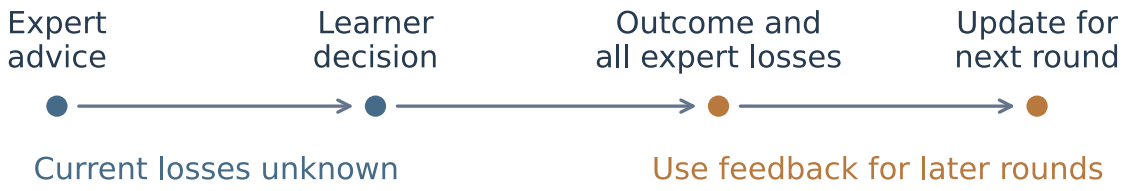
The algorithms share a simple idea. Keep a record of how the experts have performed. Use that record to make the current decision. Observe the outcome, then change the record for later decisions. The key restriction is that today's loss cannot be used to choose today's decision after the fact.

## Specify one round

**Definition: Full-information online prediction**

There are $n \geq 1$ experts and $T \geq 1$ rounds. In round $t$, the learner observes the experts' predictions and commits to its own decision before observing the current outcome. The outcome is then revealed. The learner incurs a loss, and the losses of **all** experts become available. This full feedback may be used to choose the next decision.

For binary prediction, the possible labels are 0 and 1. The zero-one loss is 0 for a correct prediction and 1 for an incorrect prediction. Thus cumulative zero-one loss is simply a mistake count. We will later allow intermediate loss values, but keep the distinction between a current decision and later feedback.




*Read from left to right. Expert advice is available before the learner decides. The current losses arrive afterward and can guide the next decision. They cannot revise the decision already made.*

"Full information" refers to the feedback after the decision. It does not mean that the learner sees future losses. If only the loss of the chosen expert were revealed, we would have a different feedback problem. All algorithms and proofs here use the complete current loss vector.

**Definition: Expert loss vector and learner loss**

Write $\ell_t(i)$ for expert $i$'s loss in round $t$. The round's expert loss vector is

$$\ell_t = (\ell_t(1), ..., \ell_t(n)).$$

The index $t$ identifies a round; the index $i$ identifies an expert. Write $L_t(\mathcal{A})$ for the learner's **scalar** loss in that round. Both are evaluated against the same revealed outcome or loss assignment.

The vector contains one entry for every expert. The learner incurs one scalar loss. For example, if three expert losses are $(0, 1, 1)$ and the learner follows the first expert, its loss is 0. It does not incur the sum 2. This distinction will matter when the learner chooses an expert at random.

## Compare with one fixed expert

After the sequence ends, add each expert's losses over all rounds. The expert with the smallest total is the **best fixed expert in hindsight**. "Fixed" means that the comparator is one expert across the whole sequence. "In hindsight" means that we choose which expert to compare with only after seeing all its losses.

**Definition: External regret**

For a scalar learner loss $L_t(\mathcal{A})$, external regret after $T$ rounds is

$$R_T = \sum_{t=1}^{T} L_t(\mathcal{A}) - \min_i \sum_{t=1}^{T} \ell_t(i).$$

The learner and every comparator use the same loss sequence. The minimum is taken after each expert's losses have been summed.

The learner does not know the best expert in advance. The point of the guarantee is to limit how much this lack of advance knowledge costs. We do not ask the learner to predict the comparator's identity correctly. We ask it to control its total loss relative to that comparator.

**Worked example: A fixed expert differs from a round-by-round choice**

On three rounds, suppose the learner losses are $(1, 0, 1)$. Two experts have losses $(0, 1, 1)$ and $(1, 0, 0)$. Their totals are already summed:

| Learner total | Expert 1 total | Expert 2 total |
|---|---|---|
| 2 | 2 | 1 |

The best fixed expert is expert 2, so $R_3 = 2 - 1 = 1$. Choosing a different zero-loss expert on each round would give total loss 0, but that is a different comparator. It is not the minimum in the definition above.

Regret need not be nonnegative. A learner that changes its decisions can sometimes outperform every single fixed expert. The term still means the same difference of cumulative losses. A regret upper bound limits how far behind the learner can be; it does not claim that the learner must remain behind.

## Read an average guarantee correctly

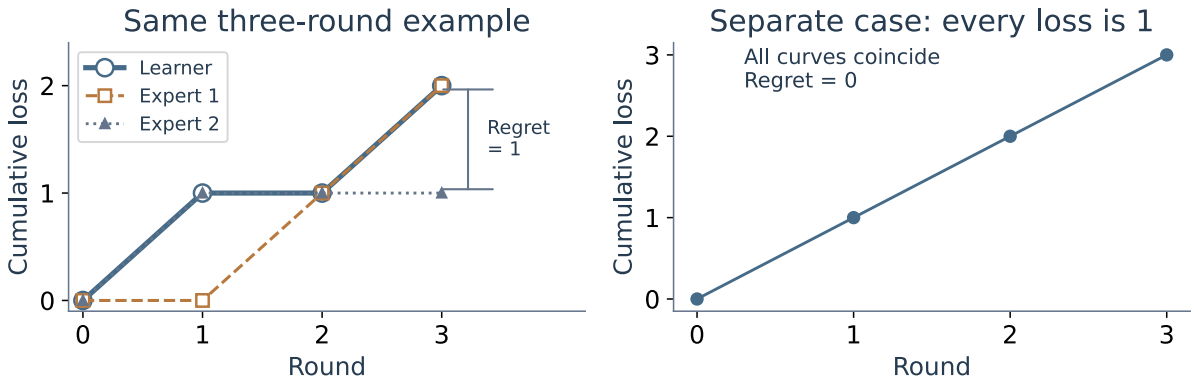
**Definition: Sublinear regret**

A regret sequence is sublinear if $R_T = o(T)$, meaning $\frac{R_T}{T}$ tends to zero as the number of rounds grows. A bound $R_T \le b(T)$ with $\frac{b(T)}{T} \to 0$ controls the learner's upper excess loss per round. This is an upper guarantee. A learner may do much better than every fixed expert. Its regret ratio can then approach a negative number instead of zero.

Dividing the regret identity by $T$ gives an exact comparison:

$$\frac{1}{T}\sum_{t=1}^{T} L_t(\mathcal{A}) = \frac{1}{T}\min_i \sum_{t=1}^{T} \ell_t(i) + \frac{R_T}{T}.$$

The first term on the right is the best fixed expert's average loss. The second is the learner's excess average loss. A small second term says little about the first one. If every expert loses one unit every round, any learner following these experts also loses one unit per round. Its regret is zero, although its absolute average loss remains one.




*Left: cumulative losses from the same three-round worked example. Expert 2 is the best fixed expert; the final gap is one. Right: a separate case in which every loss is one. All curves coincide, so regret is zero while average loss is one.*

The plot separates the size of the loss from the size of the comparison gap. The expert curves on the left are whole trajectories; we do not join the lowest pieces of different experts into a new comparator. On the right there is no gap at all, even though every cumulative loss grows.

The comparison is now precise, but we still need a decision rule. Suppose first that at least one expert is always correct. A wrong prediction then tells us which experts can no longer be perfect.

# Keep experts that have not yet been wrong

**Definition: Perfect expert**

For the horizon under consideration, a perfect expert makes no mistakes on any round. The assumption is that at least one such expert is in the original pool. Being the best expert in hindsight is weaker: the best expert may still make mistakes.

If a perfect expert exists, an expert that has already been wrong cannot be that expert. This suggests maintaining an active set of experts that have not yet made a mistake. Predict by the majority among them, then remove the active experts whose advice was wrong. This rule is called Halving.

**Definition: Halving algorithm**

Initialize the active set $S_1 = \{1, ..., n\}$. On round $t$, predict the majority binary label among experts in $S_t$; use label 0 for a tie. After observing $y_t$, retain exactly the active experts whose prediction equals $y_t$. The resulting set is $S_{t+1}$.

Removal occurs after every revealed outcome, including a round when the learner itself is correct. A minority expert can be wrong even when the learner is right. The active set records which experts have never been wrong, not merely which experts agreed with the learner.

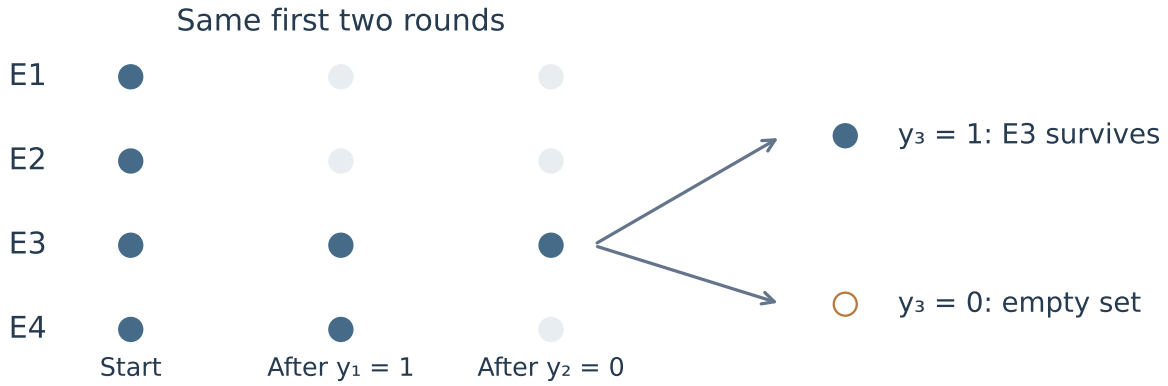
## Follow the same experts through a short prefix

We will reuse four expert IDs. The following table fixes their predictions for three rounds. The first two outcomes are 1 and 0. We temporarily leave the third outcome open so that we can see what the perfect-expert assumption changes.

| Round | Expert 1 | Expert 2 | Expert 3 | Expert 4 | Outcome |
|---|---|---|---|---|---|
| 1 | 0 | 0 | 1 | 1 | 1 |
| 2 | 0 | 1 | 0 | 1 | 0 |
| 3 | 1 | 0 | 1 | 0 | To be specified |

On round 1 all four experts are active. Their vote is tied, so Halving predicts 0 and is wrong. Experts 1 and 2 are removed. On round 2 the active experts are 3 and 4. Their vote is again tied, so the prediction is 0. This time the prediction is correct, but expert 4 is wrong and is removed. Only expert 3 remains.

If the third outcome is 1, expert 3 is correct again. It is perfect on this three-round sequence. If the third outcome is instead 0, expert 3 is wrong and the active set becomes empty. The predictions in the table have not changed; only the third outcome has changed.




*Blue dots are active experts; pale dots have been removed. The first two rounds are shared. The two final branches differ only in the third outcome. A perfect expert survives on the upper branch; the lower branch has no perfect expert.*

The first branch shows why the algorithm can be useful under its stated assumption. The second shows exactly where that assumption is needed. An empty active set supplies no majority prediction. The theorem below will prevent that event by keeping a perfect expert in the set.

## Prove a bound by counting survivors

**Result: Halving mistake bound**

Suppose at least one of the $n$ experts is perfect over the $T$ rounds. Then Halving makes at most $\log_2 n$ mistakes. Since the mistake count is an integer, it is at most $\lfloor \log_2 n \rfloor$.

Let $M$ be the learner's total mistake count. Consider a round on which the learner is wrong. At least half of the active experts voted for its predicted label, including the tied case. Those experts were wrong and are removed. Thus, on every learner mistake,

$$|S_{t+1}| \leq \frac{|S_t|}{2}.$$

On other rounds, the active set cannot increase. Starting with $n$ experts and applying these reductions gives

$$|S_{T+1}| \leq \frac{n}{2^M}.$$

This upper bound alone allows the final set to be empty. A perfect expert supplies the needed lower bound: it is never removed, so

$$|S_{T+1}| \geq 1.$$

Combining the two statements gives

$$1 \leq |S_{T+1}| \leq \frac{n}{2^M} \quad \Rightarrow \quad 2^M \leq n \quad \Rightarrow \quad M \leq \log_2 n.$$

The final logarithm counts learner mistakes, not elapsed rounds. Many correct rounds can occur without using up this mistake allowance. Nor must the remaining set contain exactly one expert; several experts may continue to give correct advice.

**Worked example: Identify the premise that is lost**

For the four-expert table, Halving's active-set sizes after the first two rounds are 2 and 1. If the third outcome is 0, the final size is 0. The learner made mistakes on rounds 1 and 3.

The shrinkage statement still holds. What fails is $|S_{T+1}| \geq 1$: there was no perfect expert on this completion. Therefore the proof cannot combine the two bounds. This does not contradict the perfect-expert theorem.

From now on, use the completion with outcomes $(1, 0, 0)$. Every expert makes at least one mistake, so the perfect-expert assumption is unavailable. We still want to compete with the best fixed expert, whose total may be small but nonzero. Removing an expert permanently after one mistake is too strong for this objective. We need a rule that reduces its influence while keeping it available.

# Replace deletion with smaller weights

Give every expert a positive weight. A larger weight gives that expert more influence on the current vote. After an expert is wrong, multiply its weight by a factor between zero and one. An expert that was correct keeps its weight. Unlike deletion, this rule leaves every expert available for later rounds.

**Definition: Deterministic weighted majority**

Choose a fixed learning rate $0 < \eta \leq \frac{1}{2}$ and initialize $w_1(i) = 1$ for every expert. On each round, add the weights voting for label 0 and for label 1. Predict the label with the greater total weight, using 0 for a tie. After observing the outcome, update every expert:

$$w_{t+1}(i) = \begin{cases} (1-\eta)w_t(i) & \text{if expert } i \text{ is wrong,} \\ w_t(i) & \text{if expert } i \text{ is correct.} \end{cases}$$

The learning rate controls the amount of reduction. With $\eta = \frac{1}{2}$, a mistaken expert keeps half its previous weight. With $\eta = \frac{1}{10}$, it keeps nine tenths. Every factor is positive, so every weight remains positive after any finite number of rounds. The vote is always defined, even when all experts have made mistakes.

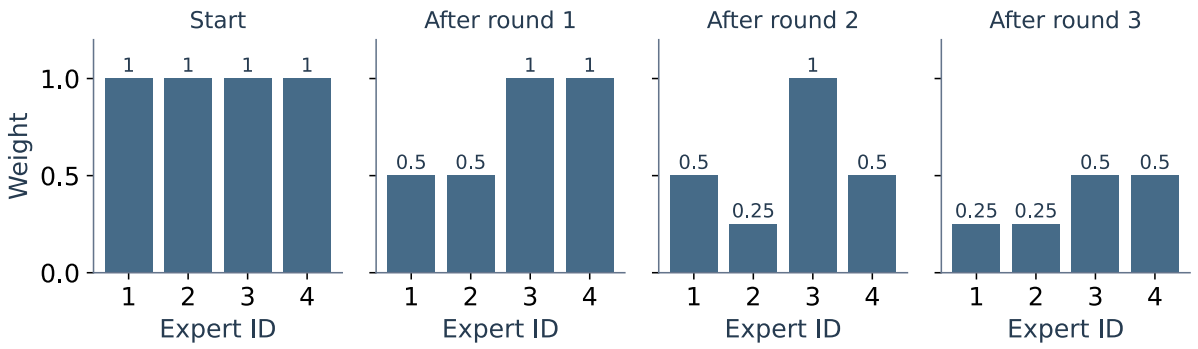
Correct experts do not gain weight in absolute terms. Their **relative** influence can increase because other experts lose weight. Also, the update uses every expert's revealed loss, not just whether the learner was wrong. The full-information feedback lets us evaluate all the advice on that round.

## Reuse the four-expert sequence

Use $\eta = \frac{1}{2}$ and the same expert predictions with outcomes $(1, 0, 0)$. The computations are supplied below. The weights shown are those available **before** each decision.

| Round | Weights before decision | Weight voting 0 | Weight voting 1 | Prediction |
|---|---|---|---|---|
| 1 | $(1, 1, 1, 1)$ | 2 | 2 | 0 |
| 2 | $\left(\frac{1}{2}, \frac{1}{2}, 1, 1\right)$ | $\frac{3}{2}$ | $\frac{3}{2}$ | 0 |
| 3 | $\left(\frac{1}{2}, \frac{1}{4}, 1, \frac{1}{2}\right)$ | $\frac{3}{4}$ | $\frac{3}{2}$ | 1 |

The learner is wrong on rounds 1 and 3, so it makes two mistakes. The expert totals are $(2, 2, 1, 1)$. Thus the best fixed expert makes one mistake, and the learner's regret on this sequence is one. Experts 3 and 4 tie as best; we do not need a unique comparator.




*These are unnormalized weights on a common vertical scale. Every mistaken expert is reduced, but none is deleted. After round 3 the weights are* $\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}\right)$*. The expert IDs and outcomes are the same as in the preceding table.*

Compare round 2 with Halving. The learner predicts correctly, but experts 2 and 4 are wrong. Their weights are reduced even though the learner incurs no loss. Keeping this distinction explicit will help us analyze the total weight.

## Relate total weight to mistakes

A weight records an expert's accumulated past mistakes. Let $M_i$ be the number of mistakes made by expert $i$ over the full horizon. Since its weight starts at one and is multiplied by $1 - \eta$ once per mistake,

$$w_{T+1}(i) = (1 - \eta)^{M_i}.$$

The index $T + 1$ refers to the state after all $T$ updates. We will consistently use this index for final weights.

Define the total weight before round $t$ by

$$\Phi_t = \sum_{i=1}^{n} w_t(i).$$

This positive scalar is called a **potential** in the proof: it is a quantity we track to connect the algorithm's changing state with its performance. Initially $\Phi_1 = n$. Total weight is not the learner's loss, and it is not a probability. We will bound it in two ways.

First, consider a learner mistake. At least half of the current weight voted for the wrong label. That part is multiplied by $1-\eta$. Even if all remaining weight stays unchanged,

$$\Phi_{t+1} \le \frac{\Phi_t}{2}(1-\eta) + \frac{\Phi_t}{2} = \Phi_t\Big(1-\frac{\eta}{2}\Big).$$

If more than half the weight was wrong, the decrease is larger. On a correct learner round, some experts may still be downweighted, but no weight increases. Hence after $M$ learner mistakes,

$$\Phi_{T+1} \le n\Big(1-\frac{\eta}{2}\Big)^M.$$

This upper bound depends on the learner's mistake count.

The second bound uses one expert. Since total weight includes every positive expert weight,

$$\Phi_{T+1} \ge w_{T+1}(i) = (1-\eta)^{M_i}$$

for every expert $i$. This lower bound depends on that expert's mistake count. A good expert prevents total weight from becoming too small. Combining these two facts will compare learner mistakes with expert mistakes.

## Make the logarithmic comparison explicit

We need two elementary inequalities. For $z \ge 0$, the exponential curve lies above its tangent at zero, giving

$$1 - z \le \exp(-z).$$

For the logarithm, we will use

$$-\log(1-x) \le x + x^2 \quad \text{for} \quad 0 \le x \le \frac{1}{2}.$$

Here and below log without a base denotes the natural logarithm. The base-two logarithm in Halving was written $\log_2$.

To verify the second inequality, put $g(x) = x + x^2 + \log(1-x)$. Its value at zero is zero. On the stated interval,

$$g'(x) = 1 + 2x - \frac{1}{1-x} = \frac{x(1-2x)}{1-x} \ge 0.$$

Therefore $g(x) \ge 0$, which is the required inequality. The restriction $x \le \frac{1}{2}$ makes the numerator nonnegative and will be used with $x = \eta$.

Apply the exponential inequality to the learner's upper bound:

$$\Phi_{T+1} \le n \exp\left(-\eta\frac{M}{2}\right).$$

Apply the logarithmic inequality to the expert's weight:

$\log(w_{T+1}(i)) = M_i \log(1-\eta) \ge -\eta(1+\eta)M_i,$

so

$$\Phi_{T+1} \ge \exp(-\eta(1+\eta)M_i).$$

Both inequalities constrain the same positive total weight. Combining them gives

$$\exp(-\eta(1+\eta)M_i) \le n \exp\left(-\eta\frac{M}{2}\right).$$

Taking natural logarithms preserves the inequality. Rearranging then gives

$$-\eta(1+\eta)M_i \le \log n - \eta\frac{M}{2},$$

$$\eta\frac{M}{2} \le \log n + \eta(1+\eta)M_i.$$

Division by the positive number $\frac{\eta}{2}$ yields the result.

**Result: Weighted-majority mistake comparison**

For binary predictions, equal initial weights and a fixed rate $0 < \eta \le \frac{1}{2}$, deterministic weighted majority satisfies

$$M \le 2(1+\eta)M_i + \frac{2\log n}{\eta}$$

for every fixed expert $i$. In particular, this holds with $M^* = \min_i M_i$, the best fixed expert's mistake count.

This theorem does not assume a perfect expert. It compares the learner with an imperfect expert through a multiplicative factor and an additive term. If $M^* = 0$, only the additive term remains. If $M^*$ grows proportionally to $T$, this bound does not force $M - M^*$ to be sublinear.

The proof also shows why the two bounds have different jobs. The upper bound alone says that many learner mistakes reduce total weight. The lower bound alone says that a good expert retains weight. Only their combination says that the learner cannot make too many mistakes while an expert makes few.

## Why deterministic prediction can still have linear regret

Consider two constant experts. Expert 1 always predicts 0; expert 2 always predicts 1. For any deterministic learner that outputs a binary label, an opponent can determine that label from the known rule and past information. It can choose the opposite outcome on each round. The learner is then wrong every time, so $M = T$.

Exactly one of the two experts is wrong on each round. Their mistake totals sum to $T$, so the better one has $M^* \le \frac{T}{2}$. Consequently,

$$R_T = M - M^* \ge \frac{T}{2}.$$

This is a linear lower bound on regret for that sequence. It applies to deterministic binary prediction, not to every possible way of combining real-valued predictions under other losses.

For weighted majority with equal initial weights and tie rule 0, the opponent's outcomes alternate $1, 0, 1, 0, \ldots$. After the first outcome, expert 1 loses weight and the learner follows expert 2. After the next outcome, expert 2 loses weight and the weights tie again. For even $T$, both experts make exactly $\frac{T}{2}$ mistakes while the learner makes $T$.

A smaller learning rate changes the weights but does not hide a deterministic decision. The opponent can still predict it. To obtain a better regret comparison, we will keep the weight update and change how the learner chooses its current prediction.

# Sample without revealing the current draw

Instead of following the label with the largest total weight, sample an expert with probability proportional to its weight. Then follow that expert's prediction. A heavier expert is more likely to be selected, but its selection is not certain. The opponent may know these probabilities without knowing the current draw.

This change helps only under a suitable information order. If the opponent sees the selected expert before choosing the loss vector, it can deliberately assign a large loss to that selection. We must specify what the opponent can know, rather than treating the word "random" as a guarantee by itself.

**Definition: Nonanticipating adversary**

A nonanticipating adversary may use the past and the learner's current public probability distribution. It fixes the current loss vector without observing the learner's current private random draw. The fixed vector is revealed only after the learner commits to its current decision. Past random draws may influence later loss vectors.

One clear way to implement this order is to publish the probabilities, let the opponent fix a hidden loss vector, draw privately, and then reveal the vector. Fixing the vector is not the same as revealing it. The learner cannot use its hidden values to select the expert.

An **oblivious** opponent fixes the whole sequence before play starts and therefore satisfies this restriction. Nonanticipation allows more adaptation: later losses can respond to earlier play. The restriction concerns access to the current draw, not a demand that the entire sequence be fixed in advance.

## Normalize weights into probabilities

**Definition: Expert sampling distribution**

For positive weights, let $\Phi_t = \sum_j w_t(j)$ and define

$$p_t(i) = \frac{w_t(i)}{\sum_j w_t(j)} = \frac{w_t(i)}{\Phi_t}.$$

Then $p_t(i) > 0$ and $\sum_i p_t(i) = 1$. Drawing $I_t$ from this distribution means that the probability of selecting expert $i$ is $p_t(i)$.

The vector $p_t$ describes the sampling rule. The random index $I_t$ describes its realized selection. They are different objects. Once the draw is made, the learner follows one expert and incurs one coordinate loss, $\ell_t(I_t)$.

**Definition: Randomized weighted majority**

Use the same positive weights and binary-loss update as deterministic weighted majority, with fixed $0 < \eta \leq \frac{1}{2}$. Before observing the current loss vector, draw expert $I_t$ privately with probabilities $p_t(i) = \frac{w_t(i)}{\Phi_t}$ and follow its prediction. After all expert losses are revealed, downweight every mistaken expert by $1 - \eta$.

Because the update uses all expert losses, it does not depend on which expert happened to be sampled. For a fixed outcome sequence, deterministic and randomized weighted majority therefore have the same weight states. Their prediction rules and learner losses differ.

For our three-round sequence, the probabilities before each decision are obtained by normalizing the earlier weight table:

| Round | Sampling probabilities | Expert loss vector | Weighted average loss |
|---|---|---|---|
| 1 | $\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right)$ | $(1, 1, 0, 0)$ | $\frac{1}{2}$ |
| 2 | $\left(\frac{1}{6}, \frac{1}{6}, \frac{1}{3}, \frac{1}{3}\right)$ | $(0, 1, 0, 1)$ | $\frac{1}{2}$ |
| 3 | $\left(\frac{2}{9}, \frac{1}{9}, \frac{4}{9}, \frac{2}{9}\right)$ | $(1, 0, 1, 0)$ | $\frac{2}{3}$ |

The last column requires an explanation. It is not a possible binary loss on a single draw. On round 1, for example, the learner's realized loss is either 0 or 1, not $\frac{1}{2}$. The fraction is its expected loss over the private draw, once the current vector and probabilities are fixed.

## Derive the scalar expected loss

For fixed probabilities and a current loss vector that does not depend on the private draw, sum over all possible selected experts:

$$E[\ell_t(I_t) \mid p_t, \ell_t, \text{pre-draw information}] = \sum_{i=1}^{n} p_t(i)\ell_t(i).$$

The conditioning means that we hold the available history, the current probability vector and the already fixed current loss vector constant. The remaining randomness is the expert draw. Expert $i$ contributes its loss multiplied by its probability of being selected.

Call the scalar weighted average $\langle p_t, \ell_t \rangle = \sum_i p_t(i)\ell_t(i)$ the **mixture loss**. The angle brackets denote the ordinary dot product. For binary losses, it equals the total probability assigned to mistaken experts. It is also their total wrong weight divided by total weight.

For the supplied sequence, the cumulative mixture loss is $\frac{1}{2} + \frac{1}{2} + \frac{2}{3} = \frac{5}{3}$. The best fixed expert's total is 1, so the mixture comparison gap is $\frac{2}{3}$. Deterministic weighted majority made two mistakes and had regret one on this same sequence. A randomized run need not realize loss $\frac{5}{3}$; that number describes an average over possible draws.

This calculation explains the difference in the potential analysis. A deterministic learner mistake tells us that at least half the weight was wrong. The mixture loss gives the **exact** fraction of wrong weight. We will use that exact fraction, rather than the half-weight lower estimate.

## Allow fractional losses without changing the information order

Binary loss only records correct or incorrect. Some decision problems assign intermediate costs. Suppose the full vector has entries in $[0,1]$. A value 0.3 now means a loss of 0.3 units; it is not necessarily a probability of a mistake. The learner still decides before observing this vector.

**Definition: Linear multiplicative weights update**

For $n \geq 1$ experts, initialize $w_1(i) = 1$ and choose a fixed rate $0 < \eta \leq \frac{1}{2}$. On each round, normalize to $p_t$, make the decision, then observe the full vector $\ell_t \in [0,1]^n$. Update every expert by

$$w_{t+1}(i) = w_t(i)(1 - \eta \ell_t(i)).$$

For mixture decisions the scalar round loss is $\langle p_t, \ell_t \rangle$. For a private sampled expert it is the realized loss $\ell_t(I_t)$, with the conditional expectation just derived.

This rule is called linear because the multiplicative factor is linear in the current loss. When the loss is zero, the factor is one. When the loss is one, it is $1 - \eta$. Intermediate losses produce intermediate factors. Thus binary randomized weighted majority is a special case of this rule.

For the stated loss and rate ranges,

$$1 - \eta \leq 1 - \eta \ell_t(i) \leq 1.$$

Every factor is at least one half, so all finite-round weights and their sum remain positive. Normalization is legal. The bound also shows that weights do not increase under nonnegative losses. An unbounded loss would not satisfy this argument and could make the linear factor negative.

**Worked example: Continue the weights with a fractional-loss round**

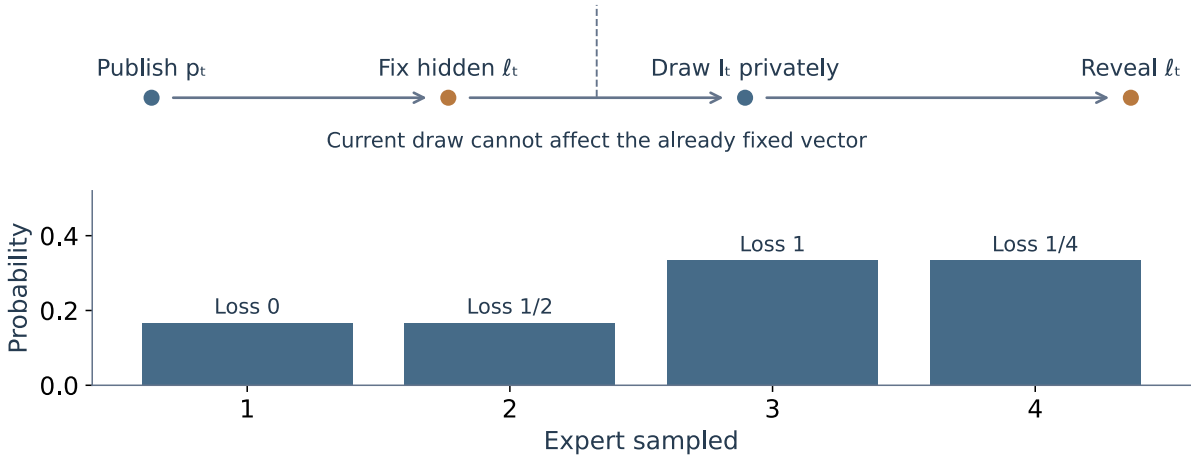
After the three binary rounds, the weights are $\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}\right)$. Keep $\eta = \frac{1}{2}$. Now change the feedback problem explicitly: round 4 supplies the bounded loss vector $\left(0, \frac{1}{2}, 1, \frac{1}{4}\right)$ rather than a binary outcome.

The pre-round probabilities are $\left(\frac{1}{6}, \frac{1}{6}, \frac{1}{3}, \frac{1}{3}\right)$. The mixture loss is

$$\frac{1}{6}0 + \frac{1}{6}\frac{1}{2} + \frac{1}{3}1 + \frac{1}{3}\frac{1}{4} = \frac{1}{2}.$$

The multiplicative factors are $\left(1, \frac{3}{4}, \frac{1}{2}, \frac{7}{8}\right)$. The next weights are $\left(\frac{1}{4}, \frac{3}{16}, \frac{1}{4}, \frac{7}{16}\right)$, with total $\frac{9}{8}$. The next probabilities are $\left(\frac{2}{9}, \frac{1}{6}, \frac{2}{9}, \frac{7}{18}\right)$.

The expert IDs and incoming weights are unchanged. The new feature is the loss domain. A sampled expert incurs one of the four listed losses, while their probability-weighted average is $\frac{1}{2}$.




*Top: the loss vector is fixed before the private draw and revealed afterward. There is no information path from the current draw back to that fixed vector. Bottom: the round-4 sampling probabilities and coordinate losses give the scalar expectation by weighting each possible loss.*

The order of the arrows is a condition for the expectation formula. Publishing a distribution does not disclose its random draw. Conversely, keeping a draw private does not help if the current vector is later chosen after observing that draw through the learner's action. Our expected-loss guarantee uses the stated nonanticipating order.

The multiplication resembles AdaBoost at a broad level, but the objects and factors differ. AdaBoost reweighted training observations with an exponential factor. Here we weight experts with $1 - \eta\ell_t(i)$ and compare sequential losses. That analogy does not prove a regret bound. We need to analyze this update on its own terms.

# Use total weight to bound the extra loss

To compare the learner with a fixed expert, first analyze the mixture loss. This scalar is determined by the current probabilities and loss vector. In this analysis set

$$L_t(\mathcal{A}) = \langle p_t, \ell_t \rangle.$$

For a fixed expert $i$, write its cumulative loss as $L_i = \sum_{t=1}^{T} \ell_t(i)$, and let $L^* = \min_i L_i$. The subscript on $L_i$ labels an expert, while $L_t(\mathcal{A})$ labels the learner's loss on one round. We will keep the sums visible where this distinction matters.

The proof uses no assumption about how favorable the sequence is. It needs the loss interval $[0, 1]$, positive equal initial weights, a fixed rate $0 < \eta \leq \frac{1}{2}$, and the stated linear update. We first prove a deterministic inequality for the mixture losses on every realized sequence. Only afterward will we interpret it for a learner that samples an expert.

## Obtain the exact one-round identity

The total weight after an update is the sum of the updated expert weights. Substitute the rule and factor out the current total:

$$\begin{aligned}\Phi_{t+1} &= \sum_i w_t(i)(1-\eta\ell_t(i)) \\ &= \Phi_t - \eta\sum_i w_t(i)\ell_t(i) \\ &= \Phi_t\left(1-\eta\sum_i p_t(i)\ell_t(i)\right) = \Phi_t(1-\eta L_t(\mathcal{A})).\end{aligned}$$

The normalization $w_t(i) = \Phi_t p_t(i)$ is what turns the weighted loss sum into a scalar learner loss. This is an equality, not yet a regret bound.

For the fractional fourth round, total weight changes from $\frac{3}{2}$ to $\frac{9}{8}$. The formula gives the same value:

$$\frac{3}{2}\left(1-\frac{1}{2}\frac{1}{2}\right) = \frac{9}{8}.$$

The numerical check shows the roles of the three factors: current total weight, learning rate and current mixture loss. The identity itself applies to any number of experts and any bounded loss vector.

A large mixture loss causes a larger proportional decrease in total weight. This provides a connection between the learner's accumulated loss and the final state. However, a total weight that decreases rapidly is not automatically a success or a failure. We still need to compare its size with what a fixed expert retains.

## Upper-bound total weight using learner loss

Because the mixture is a weighted average of numbers in $[0,1]$, its value also lies in $[0,1]$. Therefore the factor $1-\eta L_t(\mathcal{A})$ is positive. Using $1-z \le \exp(-z)$ in the exact identity gives

$$\Phi_{t+1} \le \Phi_t \exp(-\eta L_t(\mathcal{A})).$$

Apply the inequality successively from round 1 through round $T$. Since $\Phi_1 = n$,

$$\Phi_{T+1} \le n\prod_{t=1}^{T}\exp(-\eta L_t(\mathcal{A})) = n\exp\left(-\eta\sum_{t=1}^{T}L_t(\mathcal{A})\right).$$

This is the learner-dependent upper bound. The sum in the exponent is cumulative mixture loss. We have not taken an expectation over sampled decisions, and we do not need to do so to establish this inequality.

Compare this with deterministic weighted majority. There we could guarantee a factor $1-\frac{\eta}{2}$ only on learner mistakes. Here the exact decrease records the mixture loss on every round. That change is why the final comparison can avoid the earlier leading factor of two.

## Lower-bound total weight using one expert

Choose one fixed expert $i$. Its weight starts at one and accumulates all its update factors:

$$w_{T+1}(i) = \prod_{t=1}^{T}(1 - \eta\ell_t(i)).$$

Total weight includes this positive term, so $\Phi_{T+1} \geq w_{T+1}(i)$. To compare with the upper bound, we want a lower expression depending on the expert's cumulative loss $L_i$.

For binary losses, this product was exactly $(1-\eta)^{M_i}$. Fractional losses require one more inequality. For $0 \leq x \leq 1$,

$$(1-\eta)^x \leq 1 - \eta x.$$

To see the direction, consider the curve $f(x) = (1-\eta)^x$. Its second derivative is

$$f''(x) = (\log(1-\eta))^2(1-\eta)^x \geq 0.$$

Thus the curve is convex and lies below the straight chord joining its endpoints on $[0, 1]$. The endpoints are $f(0) = 1$ and $f(1) = 1 - \eta$, so that chord is $1 - \eta x$. At binary losses the curve and chord agree. At an intermediate loss the linear factor is at least the exponential expression.

Apply the inequality to every factor, with $x = \ell_t(i)$. All factors are positive, so multiplication preserves the order:

$$w_{T+1}(i) \geq \prod_{t=1}^{T}(1-\eta)^{\ell_t(i)} = (1-\eta)^{\sum_{t=1}^{T}\ell_t(i)} = (1-\eta)^{L_i}.$$

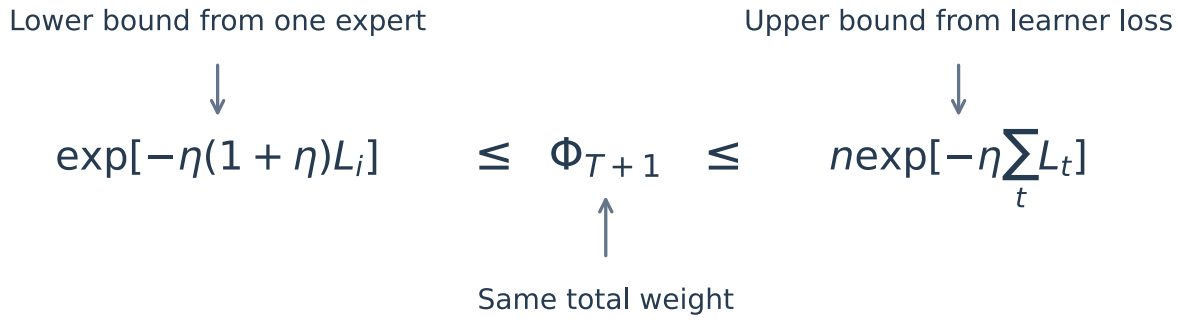
The equality adds the exponents because their base is the same. It does not replace the linear update by an exponential update. The exponential expression is a lower bound used in the proof.

The earlier logarithmic inequality gives $\log(1-\eta) \geq -\eta(1+\eta)$. Since $L_i \geq 0$, multiplying by $L_i$ keeps the direction. Exponentiating gives

$$\Phi_{T+1} \geq w_{T+1}(i) \geq (1-\eta)^{L_i} \geq \exp(-\eta(1+\eta)L_i).$$

This is the comparator-dependent lower bound. It holds for every fixed expert, including whichever expert is best after the sequence ends.

## Combine the bounds before subtracting the comparator




*Both sides constrain the same final total weight. The left side uses one fixed expert's cumulative loss. The right side uses cumulative mixture loss, with $L_t$ abbreviating $L_t(\mathcal{A})$ in this figure. The proof connects them through the middle quantity.*

The arrows identify the source of each expression. Read the inequalities from the lower bound through total weight to the upper bound. Neither side alone compares the two loss totals. Together they give

$$\exp(-\eta(1+\eta)L_i) \le n \exp\left(-\eta \sum_{t=1}^{T} L_t(\mathcal{A})\right).$$

Take logarithms, move the learner-loss term to the left, and divide by positive $\eta$:

$$-\eta(1+\eta)L_i \le \log n - \eta \sum_{t=1}^{T} L_t(\mathcal{A}),$$

$$\sum_{t=1}^{T} L_t(\mathcal{A}) \le (1+\eta)L_i + \frac{\log n}{\eta}.$$

This is a cumulative **loss** comparison. It is not yet the regret expression because the expert's loss is still present on the right.

The inequality holds for every fixed $i$, so choose an expert with total $L^*$. Subtract $L^*$ from both sides. For mixture regret this gives

$$R_T = \sum_{t=1}^{T} L_t(\mathcal{A}) - L^* \le \eta L^* + \frac{\log n}{\eta} \le \eta T + \frac{\log n}{\eta}.$$

The last step uses $L^* \le T$, which follows from one expert losing at most one unit on each of $T$ rounds. The intermediate bound retains information about a particularly good comparator; the final bound can be tuned without knowing $L^*$ in advance.

**Result: Linear MWU regret bound for bounded losses**

Let $n \ge 1$ and $T \ge 1$. Initialize $w_1(i) = 1$, use a fixed $0 < \eta \le \frac{1}{2}$, and suppose every loss $\ell_t(i)$ lies in $[0,1]$. Use the linear update and its normalized distributions. For every loss sequence,

$$\sum_{t=1}^{T} \langle p_t, \ell_t \rangle - \min_i \sum_{t=1}^{T} \ell_t(i) \le \frac{\log n}{\eta} + \eta T.$$

More precisely, the right side may use $\eta L^*$ instead of $\eta T$. No perfect-expert assumption or independent sampling of the loss sequence is required for this mixture inequality.

**Worked example: Reconstruct what each inequality contributes**

Suppose we retain only $\Phi_{T+1} \le n \exp\left(-\eta \sum_t L_t(\mathcal{A})\right)$. Can this alone prove small regret? No: it contains no comparator loss. Retaining only the lower bound also cannot control learner loss.

The connection is $w_{T+1}(i) \le \Phi_{T+1}$. It places one expert's retained weight below the same total that learner loss bounds from above. This is why we expanded the total update and the individual product separately. After combining them, the subtraction of $L^*$ turns the loss comparison into regret.

For the four-round constructed example, cumulative expert losses are $\left(2, \frac{5}{2}, 2, \frac{5}{4}\right)$ and cumulative mixture loss is $\frac{13}{6}$. The comparison gap is $\frac{13}{6} - \frac{5}{4} = \frac{11}{12}$. These are supplied observed totals for

the construction. The theorem supplies an upper bound on their gap; it does not predict that the gap equals its right side.

## Transfer the result to a sampled learner

For a learner that privately samples an expert, now take $L_t(\mathcal{A}) = \ell_t(I_t)$ in the general regret definition. Its realized cumulative loss can differ from the sum of mixture losses. We must connect their expectations rather than identify them on each run.

Under the nonanticipating condition, the conditional expectation formula holds on every round. Averaging that equality over the past and current loss-vector choices, then summing, gives

$$E\left[\sum_{t=1}^{T} \ell_t(I_t)\right] = E\left[\sum_{t=1}^{T} \langle p_t, \ell_t \rangle\right].$$

This is the law of total expectation applied round by round. It does not require the loss vectors on different rounds to be independent.

If the opponent uses past sampled actions, the future loss sequence can be random. Then the best fixed expert's cumulative loss is random too. Keep that comparator inside the expectation:

$$\begin{aligned} E[R_T] &= E\left[\sum_{t=1}^{T} \ell_t(I_t) - \min_i \sum_{t=1}^{T} \ell_t(i)\right] \\ &= E\left[\sum_{t=1}^{T} \langle p_t, \ell_t \rangle - \min_i \sum_{t=1}^{T} \ell_t(i)\right] \\ &\leq \frac{\log n}{\eta} + \eta T. \end{aligned}$$

The last line uses the mixture inequality on every realized sequence. We subtract exactly the same random comparator on both sides of the expectation equality. We do not move the minimum outside the expectation or replace the expected minimum by a minimum of expectations.

For an oblivious fixed loss sequence, the comparator is a fixed number, so it can be written outside the expectation. The more general expression above also covers adaptive nonanticipating sequences. The result bounds expected regret, not the regret of every sampled run and not a high-probability event. Binary randomized weighted majority is included by restricting every expert loss to zero or one.

## Choose a legal learning rate

The coarse bound is the sum of two terms:

$$B(\eta) = \frac{\log n}{\eta} + \eta T.$$

For $n \geq 2$, its first term decreases as $\eta$ increases, while the second increases. If the horizon $T$ is known, differentiating gives the unconstrained balance

$$B'(\eta) = -\frac{\log n}{\eta^2} + T = 0 \quad \Rightarrow \quad \eta = \sqrt{\frac{\log n}{T}}.$$

The algorithm and proof require $\eta \le \frac{1}{2}$. Therefore use

$$\eta = \min\left\{\frac{1}{2}, \sqrt{\frac{\log n}{T}}\right\}.$$

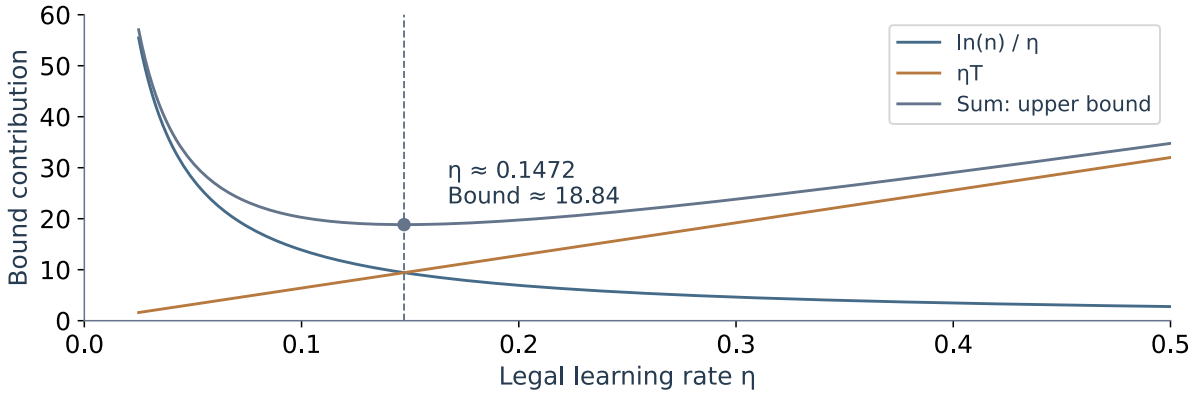
The minimum here enforces a condition of the theorem. It is not an optional adjustment made after computing the guarantee.

If $T \ge 4\log n$, the square-root choice is at most one half. Both bound contributions then equal $\sqrt{T\log n}$, so

$$R_T \le 2\sqrt{T\log n}$$

for mixture regret, and the same upper bound holds for expected sampled regret. This is the balanced regime.

If $T < 4\log n$, the legal choice is $\eta = \frac{1}{2}$, and the coarse expression instead gives $2\log n + \frac{T}{2}$. We may also use the trivial bound $R_T \le T$ because learner loss is at most $T$ and the best expert's loss is nonnegative. A valid bound can be loose; no theorem is violated when its numerical value exceeds the largest possible regret.




*For $n = 4, T = 64$, the two bound contributions balance at a legal rate near 0.1472. The vertical values are bounds, not measured losses. Only rates up to one half are displayed.*

**Worked example: Use the rate condition before the square-root formula**

The following values are supplied:

| $n$ | $T$ | Unconstrained rate | Legal rate | Available bound |
|---|---|---|---|---|
| 4 | 64 | 0.1472 | 0.1472 | $2\sqrt{64\log 4} \approx 18.84$ |
| 4 | 1 | 1.1774 | $\frac{1}{2}$ | $2\log 4 + \frac{1}{2} \approx 3.27$ |

In the second row, the unconstrained rate is illegal. Use the capped rule; the trivial regret bound 1 is sharper than 3.27. The small-horizon case does not justify substituting 1.1774 into an algorithm whose proof requires at most one half.

If $n = 1$, the learner has only one expert to use. Its loss and the comparator's loss coincide, so regret is identically zero. Keep any legal positive rate, such as $\eta = \frac{1}{2}$; do not substitute the value zero suggested by $\sqrt{\frac{\log(1)}{T}}$ into formulas containing division by $\eta$.

For fixed $n$ and sufficiently large known horizons, divide the balanced bound by $T$ to obtain $2\sqrt{\log \frac{n}{T}}$, which tends to zero. This compares a family of runs whose fixed rate is chosen for each known horizon. A single fixed positive rate in the coarse bound leaves a term $\eta$ after division by $T$. Replacing it by a different rate every round would change the proof; an unknown-horizon schedule needs separate analysis.

## Select a rule from the available assumptions

The algorithms differ in what information they use and what their guarantees compare. The final choice should follow these conditions, not the appearance of a multiplicative formula.

| Rule | Conditions used here | Guarantee established |
|---|---|---|
| Halving | Binary advice; a perfect expert exists | At most $\log_2 n$ learner mistakes |
| Deterministic weighted majority | Binary advice; fixed $0 < \eta \leq \frac{1}{2}$ | Mistake comparison with factor $2(1+\eta)$ and additive $2\frac{\log(n)}{\eta}$ |
| Randomized weighted majority | Binary losses; private current draw; nonanticipating timing | Mixture regret bound, and expected sampled regret bound |
| Linear MWU | Full loss vector in $[0,1]^n$; legal fixed rate | Mixture regret $\leq \frac{\log(n)}{\eta} + \eta T$; sampled interpretation requires nonanticipation |

> **Worked example: Choose the rule and justify its guarantee**
>
> A binary prediction task promises a perfect expert and asks for a mistake bound. Halving applies because a perfect expert keeps the active set nonempty.
>
> Now remove that promise and require a deterministic binary prediction. Weighted majority keeps all experts available and supplies its factor-style comparison. It does not supply additive sublinear regret against all sequences.
>
> If private sampling is allowed and the current loss vector is fixed without observing that draw, randomized weighted majority gives the expected-regret interpretation. If the revealed losses are fractional values in $[0,1]$, use the general linear MWU rule and its same two-sided potential argument. In each case the condition explains the choice; a best hindsight expert alone does not justify Halving.

The original question asked how to stay close to a good expert without knowing who it is. Positive weights retain imperfect experts. Private sampling changes the loss comparison when its information condition holds. The potential proof then bounds cumulative mixture loss relative to every fixed expert, allowing the best one to be chosen afterward. A small excess per round is the resulting promise; neither low absolute loss nor future population accuracy follows from that promise alone.

## Standard language to carry forward

| Term or notation | Meaning in this lesson |
|---|---|
| Full information | All current expert losses are revealed after commitment. |
| Expert loss vector $\ell_t$ | One round's losses, with one coordinate for each expert. |
| External regret $R_T$ | Cumulative learner loss minus the best fixed expert's cumulative loss on the same sequence. |
| Sublinear regret | Regret grows more slowly than $T$; a sublinear upper guarantee controls excess average loss. |
| Perfect expert | One expert that is correct on every round of the stated horizon. |
| Nonanticipating adversary | May use past information and public probabilities, but not the current private draw when fixing current losses. |
| Sampling distribution $p_t$ | Positive weights normalized to probabilities; distinct from the realized expert index $I_t$. |
| Mixture loss | The scalar dot product $\langle p_t, \ell_t \rangle$; conditional expected sampled loss under the information condition. |
| Linear MWU | Multiply each weight by $1 - \eta \ell_t(i)$ after full feedback, using the stated bounded loss and rate ranges. |